Shivam Randive

Integrated M.Sc. Physics, SVNIT Surat, India | Machine learning for materials prediction

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Research Profile

My research focuses on discovering new alloys in the Heusler family and predicting their properties for thermoelectric applications, with machine-learning predictions tested against the ground truth they replace. One thread runs through my work. At the S. N. Bose Centre I showed that graph neural networks trained on narrow data fail on unseen materials. At SVNIT this grew into two first-author manuscripts with my supervisor, Dr. Himanshu Pandey: an ML screen of 106,235 Heusler structures judged by 3,625 first-principles calculations [P1], and a correction that makes ML predictions of thermal conductivity match experiment [P2]. At NIMS, Japan, I am now extending this to new alkali- and alkaline-earth-metal Heusler alloys, predicting their Seebeck coefficient and lattice thermal conductivity with transfer-learned interatomic potentials. **Interests:** graph neural networks, interatomic potentials, uncertainty-aware screening, generative design.

Education

Integrated M.Sc. (5-year BS–MS) in Physics — **2022 – 2027 (expected)**
Sardar Vallabhbhai National Institute of Technology (SVNIT) — *Surat, India*
CGPA **8.30 / 10** (top 20% of class)

Semester Exchange Student — **Upcoming (Jan 2027)**
Indian Institute of Technology (IIT) Delhi — *New Delhi, India*

Publications

First-author manuscripts, both ready for journal submission — * corresponding author

[P1] **S. Randive**, R. Dixit, H. Pandey. **High-throughput screening of Heusler alloys for thermoelectrics combining machine-learned densities of states and first-principles transport.** Preprint: arXiv:2610.18179 (2026).

[P2] **S. Randive***, O. Randive, H. Lungaria, H. Pandey*. **Predicting the experimental lattice thermal conductivity of half-Heusler compounds by machine learning and transfer-function calibration.** For *Computational Materials Science* (2026); arXiv preprint forthcoming. Code & data: doi:10.5281/zenodo.23333947

Posters & Talks

Poster: *AI-driven discovery of Heusler alloys using generative models and electronic structure learning*, National Science Day, SVNIT Surat — Feb 2026

Research Internships

Research Intern (fully funded) · Spin Theory Group, Prof. Terumasa Tadano — **Sep – Dec 2026**
National Institute for Materials Science (NIMS) — *Tsukuba, Japan*

New alkali- and alkaline-earth Heusler thermoelectrics with pretrained AI models — ongoing

- *Idea:* build new alkali- and alkaline-earth-metal Heusler compositions combinatorially and rank them by predicted Seebeck coefficient and lattice thermal conductivity, using universal interatomic potentials (eSEN, EquiformerV3) pretrained on millions of DFT calculations.
- *Approach:* frozen transfer learning (Xiao and Tadano, *npj Comput. Mater.* 2026), comparing convolution- and attention-based equivariant networks layer by layer.

Summer Research Fellow (IASc SRFP) · Prof. Tanusri Saha-Dasgupta — **Jun – Jul 2025**
S. N. Bose National Centre for Basic Sciences — *Kolkata, India*

Do graph neural networks work on materials they have never seen? [report]

- Built a full ALIGNN pipeline and compared a broadly pretrained model with "specialists" trained only on narrow perovskite families.
- *Result:* specialists scored >92% on familiar data but only ≈48% on new double perovskites; the pretrained model kept ≈80%. *Message:* breadth of data beats narrow accuracy.

Summer Research Intern — **Jun – Jul 2024**
Indian Institute of Technology (IIT) Bhilai — *Bhilai, India*

Strain in silicon thin films for stretchable electronics [report]

- COMSOL finite-element model of a silicon film on flexible polyimide; its strain maps fed a collaborator's DFT study of how strain tunes the film's electronic properties.

Research Projects

Semester research projects in the Integrated M.Sc., Department of Physics, SVNIT, supervised by Dr. Himanshu Pandey

Finding thermoelectrics among 106,235 Heusler compounds with ML [P1] 2026

- *Problem:* transport calculations are too expensive for whole databases, so candidates are shortlisted from the cheaper density of states (DOS), often ML-predicted. This shortcut had never been tested at scale.
- *What I did:* trained 30 ALIGNN graph neural networks to predict the DOS of 54,978 structures (held-out $r = 0.90$ vs. 0.49–0.61 for simple baselines), and judged the shortlist with 3,625 first-principles transport calculations from my own Quantum ESPRESSO + BoltzTraP2 workflow.
- *Result:* a DOS-based Seebeck indicator finds the good compounds with about **4**$\times$ **fewer expensive calculations**, giving **80 candidate thermoelectrics** (61 never measured); 10 of 11 known half-Heusler thermoelectrics pass the same checks.
- *Message:* ML spectra miss most band gaps (recall 0.15) even when standard scores look good, so they can rank candidates but only a band-structure calculation can confirm a gap.

Making ML predictions of thermal conductivity match experiment [P2] 2026

- *Problem:* ML models of lattice thermal conductivity (κ_L) learn from calculated values, which I showed overestimate experiment by a median **1.69**$\times$ (29 of 34 half-Heuslers).
- *What I did:* built a source-tracked dataset of 9,495 κ_L values for 960 compounds from five databases and the literature (1,178 values read by multimodal LLMs, each physics-checked), and a simple two-parameter correction from calculation to experiment.
- *Result:* predictions of measured κ_L reach **33% median error vs. 46–47% for baselines** (cubic 18-electron compounds), close to the disagreement between two labs; the correction also cuts an independent ML force field's error from 101% to 37%. Released 10 testable predictions with 90% prediction intervals.

Simple physical descriptors as first predictors [report] Jan – Apr 2025

- Support-vector classifier for perovskite formability (≈90% accuracy); reproduced a Gaussian-process model of lattice constants for 137 half-Heusler compounds.

Next: Phono3py thermal conductivity for the [P1] candidates; crystal generation with MatterGen.

Open Data & Code

- **heusler-dos-te-screening** [P1] *(I designed and built it)*: the full screening pipeline (ML models, ranking tools and an automated DFT transport workflow) with the results of all 3,625 calculations and the 80 candidates.
- **Predicted DOS dataset** [P1] *(lead author; Zenodo, release pending)*: ML-predicted DOS for **54,978 Heusler structures** with no DFT spectrum, plus the 30 trained models.
- **half-heusler-kappa-calibration** [P2] *(lead author, archived on Zenodo)*: the 9,495-value thermal-conductivity dataset with the source of every value, the fits, the 10 predictions and the code that rebuilds every figure and number in the paper.

Fellowships & Awards

Fully funded research internship, National Institute for Materials Science (NIMS), Japan 2026
Selected for semester exchange at IIT Delhi; eligible for direct PhD admission 2026
Summer Research Fellowship (SRFP), national merit-based fellowship of the Indian Academy of Sciences, INSA and NASI 2025
Best Structural Stability and Flight Award (3rd overall), national rocketry, IIT Mandi 2025

Technical Skills

ML methods	Graph neural networks (training, fine-tuning, transfer learning); MLIP transfer learning and fine-tuning (eSEN, EquiformerV3); conformal prediction; SHAP
Classical ML	CatBoost, LightGBM, Gaussian processes, support-vector machines
ML libraries	PyTorch, DGL, fairchem, scikit-learn, NumPy, pandas, Matplotlib
First principles	VASP, Quantum ESPRESSO (incl. spin–orbit coupling), BoltzTraP2 transport, Phonopy/Phono3py
Materials data	Materials Project, AFLOW, JARVIS, OQMD, Starrydata; LLM-assisted data extraction from the literature
Computing	Python, Linux, SLURM GPU clusters, Git, MATLAB, COMSOL Multiphysics, LaTeX

Mentoring

Mentored two junior students, O. Randive and H. Lungaria, through their first research project; both are co-authors on [P2].